

```
<     print( 'led off = ' + str(led_off))
<
<     stateis = ''
<     if led_on == 6:
<         print("led on")
<         led.on()
<         stateis = "LED is ON"
<
<     if led_off == 6:
<         print("led off")
<         led.off()
<         stateis = "LED is OFF"
63c57
<     response = html % stateis
---
>     response = html % getitemp()
75,76c69
<         html,
<         led)
\ No newline at end of file
---
>         html)
\ No newline at end of file
```

--- End of diff for picowitemp.py ---

Although there are tons of sensors available for the Pico boards, I selected the ones that are inexpensive, easy to understand, and can be set up to become productive right away, helping initiate home automation with simple and short code chunks. I have just scratched the surface of the great IoT capabilities provided by the RPI Pico boards. A vast array of real-world projects is available freely on the internet. Do explore these. 

References

- MicroPython official documentation, <https://docs.micropython.org/en/latest/>
- Mpremove official documentation, <https://docs.micropython.org/en/latest/reference/mpremote.html>
- MicroPython firmware for Pico W download page, https://micropython.org/download/RPI_PICO_W/

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Pitfalls:

- Never treat Zero Trust as just a product or a single architecture; it is a long-term operating model and a set of guiding principles for workflow, system design, and operations.
- Do not try to microsegment every single resource at once; without a proper 'tuning phase' where policies are enforced while still allowing business processes to function, you will likely cause process failures.
- Ensure that Non-Person Entities (NPEs), such as service accounts and automated software agents, also have strict authentication and policy enforcement, as they can be coerced or impersonated by attackers.
- Avoid relying on weak telemetry or 'shadow IT' that isn't monitored, because if you cannot observe the current state of assets and network traffic, your Policy Engine cannot make accurate access decisions.

Transitioning to a Zero Trust Architecture is a continuous journey of risk evaluation rather than a simple technology refresh or a wholesale replacement of existing infrastructure. Most organisations will naturally operate in a hybrid mode for an indefinite period, where modern Zero Trust workflows coexist alongside legacy perimeter-based systems. To be successful, you should seek to incrementally implement these principles, starting with your highest-

value data assets while ensuring you have a deep, documented understanding of all enterprise subjects, physical and virtual assets, and critical business processes. Ultimately, by moving away from implicit trust and towards a model of continuous monitoring and granular, per-session authorisation, you can significantly reinforce your security posture and limit the potential for lateral movement within your environment. 

References

- <https://www.microsoft.com/en-us/security/blog/2024/01/25/midnight-blizzard-guidance-for-responders-on-nation-state-attack/>
- <https://doi.org/10.6028/NIST.CSWP.20>
- <https://doi.org/10.6028/NIST.SP.800-207>

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